



Is JavaScript a compiled
or interpreted language?



JavaScript is not purely interpreted and not purely compiled — it is a Just-In-Time (JIT) compiled language.



How JavaScript actually runs

When JavaScript code is executed, the engine follows these steps:

✦ Parsing

The JavaScript engine reads the code and converts it into an Abstract Syntax Tree (AST).

✦ Interpretation

The code is initially interpreted so it can start running immediately without waiting for full compilation.

✦ JIT Compilation

As the program runs, the engine identifies frequently executed (“hot”) code and compiles it into optimized machine code at runtime.

✦ Optimization & De-optimization

If assumptions change, the engine can re-optimize or de-optimize the compiled code to maintain correctness and performance.



Why JavaScript uses JIT compilation

- ✓ Fast startup time
- ✓ High runtime performance
- ✓ Efficient memory usage
- ✓ Suitable for dynamic language features

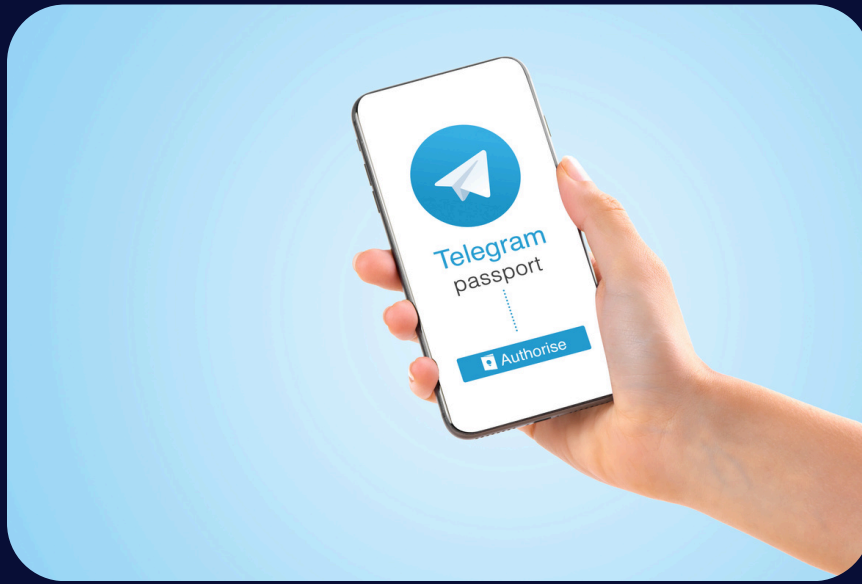


Modern JavaScript engines

- 🧠 Chrome & Node.js → V8
- 🧠 Firefox → SpiderMonkey
- 🧠 Safari → JavaScriptCore

Interview-ready conclusion

JavaScript is a JIT-compiled language where code is first interpreted and then compiled into optimized machine code at runtime for better performance.



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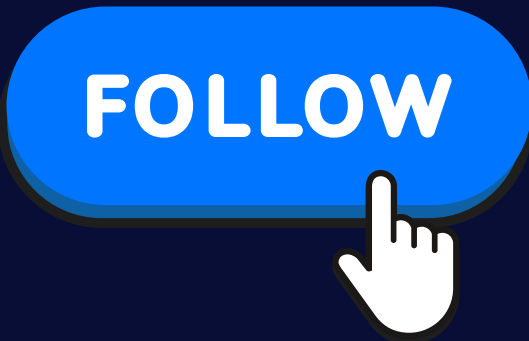


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